NGUYEN DUY TRANG

MOBILE ENGINEER

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PROFESSIONAL SUMMARY

OVERVIEW

- **Core Mobile Engineering:** Over 5 years of production experience engineering high-performance mobile applications on Flutter, Swift (iOS), and Kotlin (Android), with deep native bridge capabilities.
- **Adaptability & Problem Solving:** Hands-on production track record across Travel OTT Social Networks, Fintech, Multi-vertical POS & Self-order Kiosks, and Multimodal AI integration. Pragmatically combines native platform power with Flutter to maximize performance and accelerate time-to-market.
- **Team Collaboration & Ownership:** Highly collaborative with cross-functional partners (Product, UI/UX, Backend, QA). Proactively surfaces architectural solutions and unblocks team velocity to deliver robust releases on schedule.

KEY HIGHLIGHTS & ACHIEVEMENTS

METRICS

- **Measured Performance Optimization:** Reduced self-order Kiosk checkout time by 30%; optimized camera buffer streaming to achieve sub-2.5s AI inference latency with locked 60 FPS rendering (0 memory leaks).
- **Offline-First — Local Database:** Embedded a high-performance Rust core into Flutter via Dart FFI to directly manage the Local Database (SQLite) and execute transactional business logic, guaranteeing zero-latency response and 100% offline persistence.
- **Automated Unit Testing & Self-Testing:** Committed to rigorous engineering self-testing: designed and maintained a comprehensive suite of 277 automated unit and widget tests (100% pass rate) covering core logic and state, integrated into CI/CD.
- **Tech Sharing (SOLID & Clean Architecture):** Conducted internal tech sharing sessions on applying SOLID in Clean Architecture, helping the team write cleaner code that is easier to maintain and test.
- **1M+ Production Scale & Reliability:** Proven track record maintaining large-scale consumer applications reaching 1M+ downloads across iOS and Android, sustaining a crash-free session rate above 99.5% in production.

TECHNICAL SKILLS MATRIX

COMPETENCIES

Languages & Platforms:	Flutter & Dart, Swift (iOS), Kotlin (Android), Rust, TypeScript.
Architecture & State:	Clean Architecture, Riverpod, BLoC, MVVM, VIPER, Modularization.
Networking & Data:	Offline-First, SQLite, Realm, Secure Storage, REST APIs (Dio), WebSocket, Firebase, Dart FFI.
Performance & Advanced:	Flutter DevTools, Render Tree Optimization, AutoRoute, Gemini AI, GenUI (A2UI).
Testing & CI/CD:	Unit & Widget Testing (277 Tests), GitHub Actions, GitLab CI/CD, Fastlane.
Delivery & Monitoring:	App Store Connect, Google Play Console, Firebase App Distribution, Sentry, Crashlytics, GA4.

PROFESSIONAL EXPERIENCE

TRACK RECORD

FINVIET TECHNOLOGY • Mobile Developer (Flutter, ECO POS & Kiosk)

08/2025 – 08/2026

Core Product: ECO POS (Google Play: `vn.com.finviet.ecopos`) — Smart Retail POS, Cashier & E-Invoice Issuance system for grocery stores, retail chains, pharmacies & F&B.

Core Business Scope: Granular RBAC Permissions • Retail Sales & Multi-modal Payments (VietQR, Cards, Cash) • Product & Inventory Management (Stock-in/out) • Tax & E-Invoicing (Decree 123) • Real-time Revenue Analytics.

Tech Stack: Flutter, Dart, Clean Architecture, BLoC, Dio, SQLite/Drift, Platform Channels (Kotlin), VietQR Pay, ESC/POS Printer, Barcode Scanner, Fastlane, CI/CD.

- **Granular System RBAC:** Engineered Role-Based Access Control matrix across all store operations (owner, manager, cashier, warehouse), securing feature access and multi-branch data.
- **Retail Checkout & Multi-Modal Payments:** Built high-speed barcode checkout (cutting transaction time by 30%); integrated dynamic VietQR Pay reconciliation, POS card terminals, and cash payments.
- **Product & Inventory Management:** Developed product catalog management with variants, barcode inventory tracking, stock-in/stock-out logging, and automated low-stock threshold alerts.
- **Tax Declaration & E-Invoice Issuance:** Integrated tax reporting and certified electronic invoice (E-Invoice) issuance compliant with Decree 123 regulations and digital signatures on mobile POS terminals.
- **Revenue & Cash Flow Analytics:** Built real-time revenue dashboards, sales analytics, cashier shift reconciliation, and automated banking transaction matching.
- **Offline-First & Hardware Integration:** Architected an Offline-first pipeline (BLoC, SQLite/Drift) guaranteeing zero order loss during outages; integrated ESC/POS thermal printers and barcode scanners via Kotlin Platform Channels.
- **Performance & CI/CD Delivery:** Locked 60 FPS rendering on low-spec Android POS hardware (Sunmi, iMin, dual customer displays); automated Android/iOS build and deployment pipelines via Fastlane and GitHub Actions.

HAHALOLO • Mobile Developer (iOS & Flutter)

05/2020 – 06/2025 (5 years)

Ecosystem of 6 Products: Social Travel Network, OTT Realtime Chat, Fintech Remittance & Investing, Aquaculture Management & Store QC.

Tech Stack: iOS Swift, UIKit, VIPER, MVVM, RxSwift, WebSocket, SQLite, CallKit, Flutter, Add-to-App, Clean Architecture, Riverpod.

- **[Hahalolo — Travel Social Network (1M+ downloads)]:** Engineered core Tour, Flight, Hotel, and Community modules; built Native iOS (Swift, VIPER) and integrated Flutter modules via Add-to-App; handled responsive UI, deep linking, and sustained crash rates < 0.5% via Crashlytics.
- **[Halome — OTT Realtime Messaging]:** Built secure messaging app from scratch in Native iOS Swift (MVVM); integrated Realtime WebSockets, RxSwift, SQLite offline cache, and CallKit VoIP for lock-screen incoming calls; locked silky 60 FPS chat scrolling.
- **[Haloki — International Remittance Fintech]:** Engineered cross-border remittance app (192 countries) using Flutter; established high-security transaction pipelines, biometric authentication, KYC verification, and encrypted local storage.
- **[Hahalolo Invest — Fintech & Securities]:** Engineered stock trading application in Flutter; optimized real-time financial candlestick charting over WebSockets, portfolio analytics, and low-latency order execution flows without frame drops.
- **[Aquannets — Aquaculture Enterprise Management (Outsource)]:** Built farm management application in Flutter applying Clean Architecture + Riverpod; implemented SQLite Offline-first persistence for remote field operations with zero connectivity.
- **[SocTrip — QC & Store Release Lead]:** Led quality assurance and end-to-end testing lifecycle; drafted test case matrices, conducted regression audits, and provided release sign-offs for App Store and Google Play distributions.

PERSONAL PROJECT

FLAGSHIP ARCHITECTURE

ASTROBITE — AI Food Scanner & Nutrition Tracker

2026 - Present

Personal Project | Flutter, Dart, Riverpod, Firebase, Google Gemini AI

Tech Stack: Flutter, Dart, Riverpod, AutoRoute, Firebase Authentication, Cloud Firestore, Firebase App Check, Google Generative AI SDK (Gemini Vision & NLU), SharedPreferences, Open Food Facts API, Fastlane, GitHub Actions.

- Designed and developed a cross-platform Flutter application using Feature-First Architecture and Riverpod, implementing user authentication, food tracking, calorie and macronutrient dashboards, nutrition analytics, custom recipes, and meal planning.
- Integrated Gemini multimodal Vision to analyze meal images and extract structured nutritional data, including multi-dish decomposition, broth and topping estimation; added barcode lookup through Open Food Facts and AI-powered nutrition label parsing.
- Implemented offline-capable food logging with local caching, a pending-sync queue, duplicate handling, and Firestore-backed persistence and real-time data streams, supported by user-scoped Firestore security rules.
- Built a context-aware AI Nutrition Coach using Gemini multi-turn conversations, persisted chat history, and custom interactive Flutter components for in-chat meal logging.
- Developed Vietnamese voice-based meal logging by combining speech-to-text recognition with Gemini NLU to transform natural-language transcripts into structured nutrition records.
- Automated Android and iOS release build workflows using Fastlane and GitHub Actions, with successful release build jobs and artifact generation for both platforms.

EDUCATION

ACADEMIC

FPT POLYTECHNIC

2018 - 2021

Major: Software Engineering (Mobile Application Development)

LANGUAGES

PROFICIENCY

English: Reads and understands technical documentation, technical communication.